

Invariant Closure Principle: Positive Forms, Boundary Leakage, and Closure Certificates

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Abstract

We isolate the formal object that was missing from the cross-domain closure diagnostic: the *closure certificate*. Many hard problems share a common shape — a finite generative rule, indefinite repetition, and possible escape at a boundary at infinity. We formalise a generated system (X, T) relative to a positive form B and a boundary completion ∂X , classify its behaviour into closure modes, and define a closure certificate as a *finitely checkable* object proving that boundary leakage is controlled under the repetition limit. We give complete proofs that, for linear systems, the three principal modes admit *exact* certificate criteria (dissipative $\Leftrightarrow$ a Lyapunov form, isometric $\Leftrightarrow$ an invariant inner product, self-adjoint $\Leftrightarrow$ symmetrizability), thereby recovering the discrete Lyapunov stability theorem and the detailed-balance self-adjointness of reversible Markov chains as instances. We then prove a self-contained number-theoretic proposition: the only positive-integer 1-cycle of the $3n+1$ shortcut map is the trivial one (recovering Steiner 1977 via the certificate route). Finally we map four open problems (Collatz, Riemann, Navier–Stokes, Yang–Mills) to the *exact missing certificate* each requires. Every claim is graded [PROVEN]/[DIAGNOSTIC]/[CONJECTURAL]/[OPEN]. **We do not prove, partially prove, or reduce any open problem**; the contribution is a precise vocabulary and a catalogue separating solved certificates from missing ones.

1 Generated systems, boundaries, positive forms

Definition 1 (Generated system). A *generated system* is a pair $G = (X, T)$ where X is a state space and $T: X \rightarrow X$ a transformation. Its orbits are the sequences $(T^n x)_{n \geq 0}$. For the linear theory we take $X = \mathbb{R}^d$ and $T \in \mathbb{R}^{d \times d}$; for the arithmetic theory X is a set of integers and T a piecewise-affine map.

Definition 2 (Boundary completion). A *boundary completion* is $\overline{X} = X \cup \partial X$, where ∂X collects the limiting behaviours that repetition can approach: divergence to infinity, a singularity, a non-returning limit, or an unresolved accumulation point. “Closure” is a statement about whether orbits stay controlled relative to ∂X as $n \rightarrow \infty$.

Definition 3 (Positive form). A *positive form* on $\mathbb{R}^d$ is a symmetric matrix $B = B^\top \succ 0$. It supplies the inner product $\langle x, y \rangle_B = x^\top B y$ used to measure size, energy, phase, probability, trace, or coherence. (Semidefinite forms $B \succeq 0$ are admitted where stated.)

Definition 4 (Closure modes). For a linear $G = (\mathbb{R}^d, T)$ and a positive form B , define

$$\begin{aligned} \text{SELF_ADJOINT (reflective): } & BT = T^\top B, \\ \text{ISOMETRIC (phase): } & T^\top BT = B, \\ \text{DISSIPATIVE (monotone): } & T^\top BT \preceq B, \\ \text{CRITICAL: } & T^\top BT - B \text{ has unresolved sign on } \partial X, \\ \text{ESCAPING: } & \text{no } B \succ 0 \text{ controls } \{T^n\}. \end{aligned}$$

Definition 5 (Boundary leakage and capacity). The *boundary leakage* of (X, T) relative to B is the largest growth the form fails to suppress in one step,

$$L_B(T) = \lambda_{\max}(T^\top BT - B),$$

and the *closure capacity* is $Q_B(T) = -L_B(T)$ (containment minus leakage). Thus $Q_B \geq 0 \Leftrightarrow T^\top BT \preceq B$ (dissipative/isometric); $Q_B < 0$ signals escape in the B -norm.

Definition 6 (Closure certificate). A *closure certificate* for $G = (X, T)$ is a finitely specified, finitely checkable object $\mathcal{C}$ that proves $L_B(T) \leq 0$ (equivalently $Q_B \geq 0$), or an exact equality/inequality among $\{BT, T^\top B, T^\top BT, B\}$, under the relevant repetition limit. The defining feature is that $\mathcal{C}$ is finite while the statement it certifies (“all orbits stay controlled as $n \rightarrow \infty$ ”) is infinitary. A *control* is a companion system known to lack closure; a certificate scheme is accepted only if it rejects the controls.

Remark. The whole programme reduces to one question: *which domains admit a finite certificate, and what is it?* The hardness of an open problem is precisely the absence of a known finite certificate for its (provably correct) closure mode.

2 Certificate criteria for linear systems (proved)

Lemma 1 (Dissipative certificate = Lyapunov form). [PROVEN] Let $T \in \mathbb{R}^{d \times d}$. There exists $B \succ 0$ with $T^\top BT - B \prec 0$ (a strict dissipative certificate) if and only if $\rho(T) < 1$. In that case $B = \sum_{k \geq 0} (T^\top)^k T^k$ converges, is positive definite, and satisfies $T^\top BT - B = -I$.

Proof. ($\Leftarrow$) If $\rho(T) < 1$ then $\|T^k\| \leq Cr^k$ for some $r < 1$, so $B = \sum_{k \geq 0} (T^\top)^k T^k$ converges. B is symmetric and $x^\top Bx = \sum_{k \geq 0} \|T^k x\|^2 \geq \|x\|^2 > 0$ for $x \neq 0$, hence $B \succ 0$. Re-indexing, $T^\top BT = \sum_{k \geq 0} (T^\top)^{k+1} T^{k+1} = B - I$, so $T^\top BT - B = -I \prec 0$.

($\Rightarrow$) Suppose $B \succ 0$ and $T^\top BT - B = -Q$ with $Q \succ 0$. Extend B Hermitianly. Let $Tv = \lambda v$, $v \neq 0$ (complex). Then $v^* T^\top BT v = |\lambda|^2 v^* B v$ and also $= v^* B v - v^* Q v$, so $(1 - |\lambda|^2) v^* B v = v^* Q v > 0$. Since $v^* B v > 0$, we get $|\lambda| < 1$; as λ was arbitrary, $\rho(T) < 1$. $\square$

Remark. Lemma 1 is the discrete-time Lyapunov stability theorem. The certificate is the finite matrix B (an $O(d^2)$ object) and the finite check $T^\top BT - B \prec 0$; it certifies the infinitary statement $T^n \rightarrow 0$.

Lemma 2 (Isometric certificate = invariant inner product). [PROVEN] For $T \in \mathbb{R}^{d \times d}$ the following are equivalent: (i) there exists $B \succ 0$ with $T^\top BT = B$; (ii) $\{T^n\}_{n \geq 0}$ is bounded and T is invertible. When T has finite order m ($T^m = I$), $B = \frac{1}{m} \sum_{k=0}^{m-1} (T^\top)^k T^k$ is an exact certificate; in general a Cesàro limit of these averages is.

Proof. (i) $\Rightarrow$ (ii): $T^\top BT = B$ means T preserves $\langle \cdot, \cdot \rangle_B$, so $\|T^n x\|_B = \|x\|_B$ for all n ; equivalence of norms gives $\{T^n\}$ bounded, and an isometry of a finite-dimensional inner-product space is invertible. (ii) $\Rightarrow$ (i): boundedness gives $M = \sup_n \|T^n\| < \infty$. The averages $B_N = \frac{1}{N} \sum_{k=0}^{N-1} (T^\top)^k T^k$ are symmetric PSD with $\|B_N\| \leq M^2$, so a subsequence converges to some $B \succeq 0$. Telescoping, $\|T^\top B_N T - B_N\| = \frac{1}{N} \|(T^\top)^N T^N - I\| \leq (M^2 + 1)/N \rightarrow 0$, hence $T^\top BT = B$. Finally $\|x\|^2 = \|T^{-k} T^k x\|^2 \leq M^2 \|T^k x\|^2$ gives $x^\top B_N x \geq \|x\|^2/M^2$, so $B \succeq M^{-2} I \succ 0$. For $T^m = I$ the average is stationary and exact. $\square$

Lemma 3 (Self-adjoint certificate = symmetrizability). *[PROVEN] There exists $B \succ 0$ with $BT = T^\top B$ if and only if T is diagonalizable with real spectrum (i.e. T is symmetrizable).*

Proof. ($\Leftarrow$) Write $T = SDS^{-1}$ with D real diagonal. Put $B = (S^{-1})^\top S^{-1} \succ 0$. Then $BT = (S^{-1})^\top S^{-1} SDS^{-1} = (S^{-1})^\top DS^{-1}$, which is symmetric since D is; hence $BT = (BT)^\top = T^\top B$. ($\Rightarrow$) Given $B \succ 0$ with $BT = T^\top B$, set $M = B^{1/2} T B^{-1/2}$. Then $M^\top = B^{-1/2} T^\top B^{1/2} = B^{-1/2} (BT B^{-1}) B^{1/2} = B^{1/2} T B^{-1/2} = M$, so M is symmetric: it has real spectrum and is orthogonally diagonalizable. As $T = B^{-1/2} M B^{1/2}$ is similar to M , T is diagonalizable with the same (real) spectrum. $\square$

Corollary 1 (Reversible Markov chains carry a self-adjoint certificate). *[PROVEN] Let P be an irreducible stochastic matrix with stationary measure $\pi > 0$ satisfying detailed balance $\pi_i P_{ij} = \pi_j P_{ji}$. Then $B = \text{diag}(\pi)$ is a self-adjoint certificate ($BP = P^\top B$); equivalently P is self-adjoint on $\ell^2(\pi)$ and has real spectrum. On the mean-zero subspace $\mathbf{1}^\perp$, ergodicity gives $\rho(P|_{\mathbf{1}^\perp}) < 1$, so $\text{diag}(\pi)$ is also a dissipative certificate there (Lemma 1): the chain closes onto π .*

Proof. $BP = P^\top B$ is exactly detailed balance written in matrix form; $B \succ 0$ since $\pi > 0$. Lemma 3 gives real spectrum. Irreducibility/apperiodicity place all non-Perron eigenvalues strictly inside the unit disc, so the restriction to $\mathbf{1}^\perp$ is a strict contraction and Lemma 1 applies. $\square$

Remark (The three modes are genuinely distinct certificates). A rotation by $2\pi/k$ admits an isometric certificate (Lemma 2) but *no* self-adjoint one (its spectrum is off the real axis), confirming the hypothesis correction from the prior work order: phase closure is isometric, not self-adjoint. A defective matrix with $\rho = 1$ (a Jordan block) admits *none* of the three: it is the linear archetype of CRITICAL.

3 An arithmetic certificate, proved

We now leave the linear setting for a finite Diophantine certificate.

Definition 7 (q -shortcut map). For odd $q \geq 3$ let $T_q(n) = (qn + 1)/2^{a(n)}$ on odd $n \geq 1$, where $a(n) = v_2(qn + 1)$ is the 2-adic valuation. A k -cycle is a tuple of odd $n_1, \dots, n_k$ with $T_q(n_i) = n_{i+1}$, indices mod k , valuations $a_1, \dots, a_k$, total $A = \sum a_i$.

Lemma 4 (Cycle equation = a finite closure certificate). *[PROVEN] A k -cycle of T_q with valuations $a_1, \dots, a_k$ exists only if, writing $s_0 = 0$, $s_j = a_1 + \dots + a_j$, $A = s_k$,*

$$n_1(2^A - q^k) = \sum_{i=1}^k q^{k-i} 2^{s_{i-1}} > 0, \quad \text{hence} \quad 2^A > q^k.$$

Proof. From $2^{a_i} n_{i+1} = q n_i + 1$, induction gives $n_{j+1} = (q^j n_1 + \sum_{i=1}^j q^{j-i} 2^{s_{i-1}})/2^{s_j}$. Setting $n_{k+1} = n_1$ and multiplying by 2^A yields $n_1 2^A = q^k n_1 + \sum_{i=1}^k q^{k-i} 2^{s_{i-1}}$, i.e. the displayed identity. The right side is a sum of positive terms, and $n_1 > 0$, so $2^A - q^k > 0$. $\square$

Proposition 1 (Only the trivial 1-cycle for $3n+1$). *[PROVEN] The unique positive-integer 1-cycle of T_3 is $n = 1$ (the cycle $1 \mapsto 1$). Consequently $3n+1$ has no nontrivial fixed cycle of length one.*

Proof. For $k = 1$, Lemma 4 gives $n_1(2^a - 3) = 1$ with $a = a_1 \geq 1$. Thus $2^a - 3$ is a positive divisor of 1, forcing $2^a - 3 = 1$, i.e. $2^a = 4$, $a = 2$, and $n_1 = 1$. The orbit is $T_3(1) = (3 \cdot 1 + 1)/2^2 = 1$. No other positive integer solves the equation. $\square$

Remark. Proposition 1 recovers Steiner’s 1977 “no nontrivial circuit” result for the simplest class, here as a one-line consequence of the closure certificate of Lemma 4. The certificate is finite (one Diophantine equation); it eliminates an *infinite* family of would-be cycles.

Corollary 2 (No small nontrivial cycles, verified). *[DIAGNOSTIC] An exhaustive check of Lemma 4 over all valuation compositions with $k \leq 6$, $1 \leq a_i \leq 4$, $2^A > 3^k$ (script *examples/*) yields no positive odd integer n_1 generating a consistent cycle other than $n = 1$. This is a finite verified statement, not a proof for all k (cf. Simons–de Weger for the known unconditional bounds).*

4 Certificate catalogue and open-problem mapping

Domain	Closure mode	Certificate	Status
Finite cycles / permutations	isometric	invariant inner product (Lem. 2)	[PROVEN]
Contractions / stable linear	dissipative	Lyapunov form (Lem. 1)	[PROVEN]
Reversible Markov chains	self-adjoint	$\text{diag}(\pi)$ (Cor. 1)	[PROVEN]
Ergodic Markov chains	dissipative on $\mathbf{1}^\perp$	spectral gap	[PROVEN]
Poincaré (3-manifolds)	dissipative	Perelman $\mathcal{W}$ -entropy monotonicity	[PROVEN](Perelman)
$3n+1$ one-cycles	—	cycle equation (Prop. 1)	[PROVEN]
Collatz (full)	dissipative/mixed	no admissible infinite non-closing braid	[OPEN]
Riemann Hypothesis	self-adjoint/positive	Weil positivity of the explicit form	[OPEN]
Navier–Stokes (3D)	dissipative/critical	coercive high-frequency norm control	[OPEN]
Yang–Mills mass gap	positive + gap	reflection positivity + spectral floor	[OPEN]

The open rows share one structure: the closure *mode* is known and correct, but the *certificate* is either infinite-dimensional or not finitely reducible by any known means.

- **Collatz.** [OPEN] Mode DISSIPATIVE-on-average (mean drift $\overline{Q} = 2 \ln 2 - \ln 3 > 0$, [DIAGNOSTIC]). Missing certificate: a finite obstruction excluding *every* infinite trajectory with persistently non-positive cumulative capacity. The cycle certificate (Lem. 4) handles *periodic* escape; the *aperiodic* unbounded braid is uncertified.
- **Riemann.** [OPEN] Mode positive form: RH $\Leftrightarrow$ the Weil explicit-formula quadratic form is positive semidefinite. The finite truncations are PSD with a near-null mode ([DIAGNOSTIC]); no finite basis certifies the limit. This is the Connes–Consani arithmetic-site programme. (The positivity form is even insensitive to prime *shuffling*, which is detected separately by the explicit-formula balance — underlining that the missing object is a positivity *certificate*, not a correlation.)
- **Navier–Stokes.** [OPEN] Mode dissipative vs. critical: a dyadic-shell caricature reproduces the viscosity-driven capacity trend ([DIAGNOSTIC]), but in 3D the controlled energy norm is *supercritical*, so L_B has unresolved sign — the CRITICAL archetype. Missing certificate: a coercive norm that dominates the nonlinear transfer uniformly.

- **Yang–Mills.** [OPEN] Osterwalder–Schrader reflection positivity *is* a positive-form certificate; the open content is constructing the continuum measure and a strictly positive spectral floor (the gap).

5 Non-overclaim boundary

1. This note proves the linear certificate criteria (Lemmas 1–3), the Markov corollary, the cycle equation, and the 1-cycle proposition. [PROVEN].
2. It **does not** prove, partially prove, or reduce Collatz, RH, Navier–Stokes, or Yang–Mills. For each it states the correct closure mode and the *exact missing certificate*; supplying that certificate is the open problem itself.
3. “Closure mode” is a classification; a solution requires *constructing* the domain-native certificate, which for the open rows is not finitely available by any method known to us.
4. The cross-domain language is an organising and search tool. Where a diagnostic trend (e.g. Collatz drift, NS capacity) is reported, it is graded [DIAGNOSTIC] and is not a certificate.
5. No claim is made about physics, consciousness, E_8 as “the universe”, or a “theory of everything”. The objects here are matrices, measures, and integers.

One-sentence statement

Many hard problems are not about infinity itself but about whether a finite generative rule admits a finite certificate—a Lyapunov form, an invariant inner product, a symmetrizer, a Diophantine obstruction, or a positivity proof—that controls its behaviour at the boundary of indefinite repetition; this note formalises that object and separates the certificates we have from the ones we lack.